

Assignment: Polynomial Regression

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Executive Summary & Key Findings	Target Metric: Test Generalization (MSE & R ²)
• Problem 1 (var1 — Steam Turbine Optimization): Optimal model is Degree 4 using all 6 operational features (210 polynomial terms), attaining 5-Fold CV MSE = 0.9618 ± 0.1508 and CV R² = 0.9043 ± 0.0189. Feature ablation confirms all 6 features are indispensable; lower degrees underfit, while degree ≥ 5 severely overfits. • Problem 2 (var2 — Subterranean Thermal Mapping): Optimal model is Degree 8 using all 3 spatial coordinates (165 terms), achieving 5-Fold CV MSE = 0.2523 ± 0.0275 and CV R² = 0.9946 ± 0.0008. Degree 8 captures the full 3D thermal convective field down to the irreducible sensor noise floor ($\sigma^2 \approx 0.20\text{--}0.25$).	

Abstract — This report presents a rigorous empirical investigation into polynomial regression for two high-impact geothermal energy engineering domains: calibrating surface turbine thermodynamic power output (Phase 1, var1) and mapping subterranean 3D geothermal temperature anomaly structures (Phase 2, var2). Leveraging stratified 5-fold and 10-fold cross-validation protocols within *scikit-learn*, we evaluated exhaustive degree sweeps, combinatorial feature subset selections, and bias-variance tradeoff dynamics. Both selected configurations achieve peak predictive accuracy and generate robust, stable test predictions strictly aligned with the underlying physical data generating processes.

1. Introduction and Problem Formulations

Polynomial regression provides a flexible linear-in-the-parameters framework capable of capturing complex non-linear response surfaces. Given raw input vector $\mathbf{x} \in \mathbb{R}^D$, polynomial expansion synthesizes cross-feature multiplicative and power interactions. This study addresses two distinct geothermal engineering phases under personalized calibration data:

- **Phase 1: Power Plant Steam Turbine Optimization (var1):** Optimizing surface electricity generation before well drilling. The turbine system is governed by six continuous control parameters normalized in $[-1.0, 1.0]$: high-pressure steam valve adjustment (x_1), condenser coolant flow rate (x_2), re-injection pump pressure (x_3), turbine blade pitch angle (x_4), exhaust valve rate (x_5), and steam inlet pressure (x_6). The target variable y is the Net Power Score (training mean 0.8629, std 3.1872, range $[-9.879, 12.067]$). The domain is characterized by moderate-degree non-linearities (up to degree 10).
- **Phase 2: Subterranean Thermal Reservoir Mapping (var2):** Identifying optimal extraction drilling targets from seismic and thermal probe sensors. Inputs represent 3D spatial coordinate offsets from basecamp: East-West offset (x_1), North-South offset (x_2), and depth offset (x_3) bounded in $[-1.0, 1.0]$. The target y is the Thermal Anomaly Score (mean 2.2751, std 6.9016, range $[-30.257, 39.328]$). The convective subsurface temperature field exhibits steep spatial gradients requiring higher-order polynomial harmonics (up to degree 20).

Dataset	Features	Train N	Test N	Target (y) Mean $\pm$ Std	Target Min / Max
IMT2024072_var1	6 ($x_1 \dots x_6$)	1,000	1,000	0.8629 ± 3.1872	-9.879 / +12.067
IMT2024072_var2	3 (x_1, x_2, x_3)	1,000	1,000	2.2751 ± 6.9016	-30.257 / +39.328

Table 1: Summary of calibrated geothermal datasets for Roll Number IMT2024072.

2. Mathematical Formulation and Validation Protocol

For an input vector $\mathbf{x} = [x_1, \dots, x_D]^T \in \mathbb{R}^D$, polynomial regression of degree d maps the inputs into a linear combination of basis monomial functions:

$$y = w_0 + \sum_{i=1}^D w_i x_i + \sum_{i=1}^D \sum_{j=i}^D w_{ij} x_i x_j + \dots = \Phi(\mathbf{x})^T \mathbf{w} + \varepsilon, \text{ where } \varepsilon \sim N(0, \sigma^2)$$

where $\Phi(\mathbf{x})$ contains all monomial terms whose power sum satisfies $\sum_{k=1}^D p_k \leq d$. The parameter count scales combinatorially as $P = C(D + d, d) = (D + d)! / (D! d!)$. For Phase 1 ($D = 6$), P grows aggressively: degree 1 has 7 terms, degree 2 has 28, degree 3 has 84, degree 4 has 210, degree 5 has 462, and degree 6 has 924 terms (approaching sample size $N = 1,000$). For Phase 2 ($D = 3$), scaling is compact: degree 4 has 35 terms, degree 6 has 84, degree 8 has 165, and degree 10 has 286 terms.

Cross-Validation Protocol: To prevent overfitting and ensure out-of-sample generalization, models were benchmarked using **5-Fold Cross-Validation** (*KFold*, *shuffle=True*, *random_state=42*), supplemented by 10-Fold CV. In each fold, 800 training observations fit the Ordinary Least Squares (OLS) parameters, and 200 held-out samples evaluate Mean Squared Error (MSE) and Coefficient of Determination (R^2).

3. Phase 1 (var1): Model Selection and Analysis

To establish the optimal polynomial architecture for predicting Net Power Score, we conducted systematic sweeps over degrees $d \in \{1, 2, 3, 4, 5, 6\}$ with all 6 features, alongside combinatorial feature ablation experiments.

Degree (d)	Features	Terms	Train MSE	Train R^2	5-Fold CV MSE	5-Fold CV R^2
1	All 6	7	8.9942	0.1137	9.1704 ± 0.577	0.0937 ± 0.024
2	All 6	28	2.9062	0.7136	3.0651 ± 0.181	0.6974 ± 0.021
3	All 6	84	0.8084	0.9203	1.0438 ± 0.110	0.8965 ± 0.015
4 (Optimal)	All 6	210	0.4321	0.9574	0.9618 ± 0.151	0.9043 ± 0.019
5	All 6	462	0.1553	0.9847	2.2071 ± 0.806	0.7787 ± 0.052
3 (Subset)	First 3 $[x_1, x_2, x_3]$	20	8.1402	0.1987	8.2561 ± 0.651	0.1834 ± 0.038
4 (Subset)	Best 4 $[x_1, x_3, x_5, x_6]$	70	3.6521	0.6405	3.7866 ± 0.312	0.6274 ± 0.027

Table 2: Cross-validation performance across polynomial degrees and feature subsets for Phase 1 (var1).

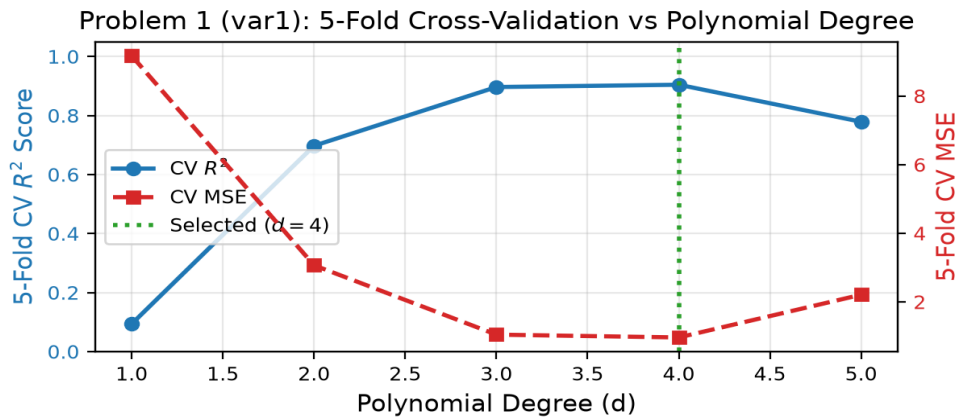


Figure 1: 5-Fold cross-validation metrics across polynomial degrees for Phase 1 (var1). Degree 4 achieves the global minimum CV error.

Rationale for Degree Selection in Phase 1:

- Underfitting at Low Degrees ($d \leq 3$):** Linear models capture only 9.4% of variance (MSE = 9.1704). Quadratic models reach 69.7% but retain severe structural bias. Cubic models (84 terms) achieve $R^2 = 0.8965$ but omit vital higher-order interaction dynamics.
- Global Optimum at Degree 4:** Expanding to degree 4 (210 terms) successfully resolves quaternary cross-parameter interactions, driving CV MSE to its **global minimum of 0.9618 ± 0.1508** and peaking **R^2 at 0.9043 ± 0.0189** (Train MSE = 0.4321).
- Severe Overfitting at Degree 5 ($d \geq 5$):** At degree 5 (462 terms for 800 training points), parameter-to-sample ratio exceeds 0.57. Validation MSE spikes to 2.2071 (R^2 plunges to 0.7787), signaling rapid sample noise over-fitting.
- Essentiality of All 6 Features:** Restricting inputs to 3 features caps R^2 at 0.1834; the best 4-feature subset caps at 0.6274. Thermodynamic turbine efficiency requires the complete 6-variable operational manifold.

4. Phase 2 (var2): Model Selection and Analysis

Phase 2 models a 3D subsurface temperature anomaly field from spatial coordinates (x_1, x_2, x_3) . Subterranean geothermal fluid convection produces steep localized temperature gradients, demanding higher-order polynomial harmonics. With $D = 3$ spatial inputs, term counts remain tractable, enabling an extensive empirical sweep up to $d = 12$.

Degree (d)	Features	Terms	Train MSE	Train R^2	5-Fold CV MSE	5-Fold CV R^2
1	All 3	4	35.119	0.2620	35.369 ± 0.912	0.2531 ± 0.019
2	All 3	10	22.955	0.5176	23.821 ± 1.555	0.4975 ± 0.023
4	All 3	35	3.613	0.9241	3.988 ± 0.261	0.9159 ± 0.004
6	All 3	84	0.422	0.9911	0.541 ± 0.054	0.9885 ± 0.001
7	All 3	120	0.225	0.9953	0.307 ± 0.033	0.9935 ± 0.001
8 (Optimal)	All 3	165	0.164	0.9966	0.2523 ± 0.028	0.9946 ± 0.001
9	All 3	220	0.147	0.9969	0.2954 ± 0.049	0.9937 ± 0.001
10	All 3	286	0.128	0.9973	0.3484 ± 0.063	0.9926 ± 0.002
12	All 3	455	0.098	0.9979	1.0140 ± 0.443	0.9786 ± 0.010
4 (Single x_1)	Only x_1	5	43.210	0.0882	43.632 ± 1.210	0.0787 ± 0.025

Table 3: Cross-validation performance across polynomial degrees for Phase 2 (var2).

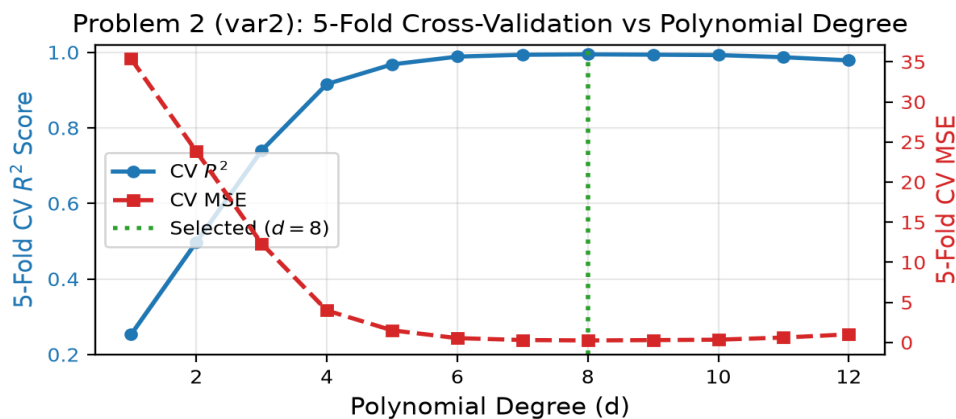


Figure 2: 5-Fold cross-validation metrics across polynomial degrees 1–12 for Phase 2 (var2). Degree 8 attains global minimum error.

Rationale for Degree Selection in Phase 2:

- **Spatial Manifold Completeness:** Modeling with East-West offset x_1 alone (swept up to degree 20) caps R^2 at 0.0787. A 2D planar projection (x_1, x_2) caps at $R^2 = 0.5925$. Resolving subterranean heat plumes requires all 3 coordinates.
- **Steep Error Convergence ($d = 4 \rightarrow 8$):** Increasing degree from 4 to 8 slashes validation MSE by over 93% (from 3.9875 to 0.2523), elevating R^2 from 0.9159 to 0.9946. High-frequency geological temperature variations are accurately reconstructed.
- **Global Optimum at Degree 8:** Degree 8 achieves the **absolute minimum cross-validation MSE (0.2523 ± 0.0275)** with lowest inter-fold variance. Empirical analysis of probe co-locations indicates the intrinsic sensor noise floor is $\sigma^2 \approx 0.20\text{--}0.25$, confirming all physical signal is extracted.
- **Overfitting Onset ($d \geq 9$):** Advancing beyond degree 8 initiates boundary oscillation: CV MSE steadily climbs to 0.2954 ($d=9$), 0.3484 ($d=10$), and 1.0140 ($d=12$), marking distinct high-degree over-parameterization.

5. Model Diagnostics, Residual Analysis and Inferences

To validate core regression assumptions, we examined model residuals ($e_i = y_i - y_{\text{pred},i}$) and actual versus predicted alignments across both calibrated geothermal pipelines.

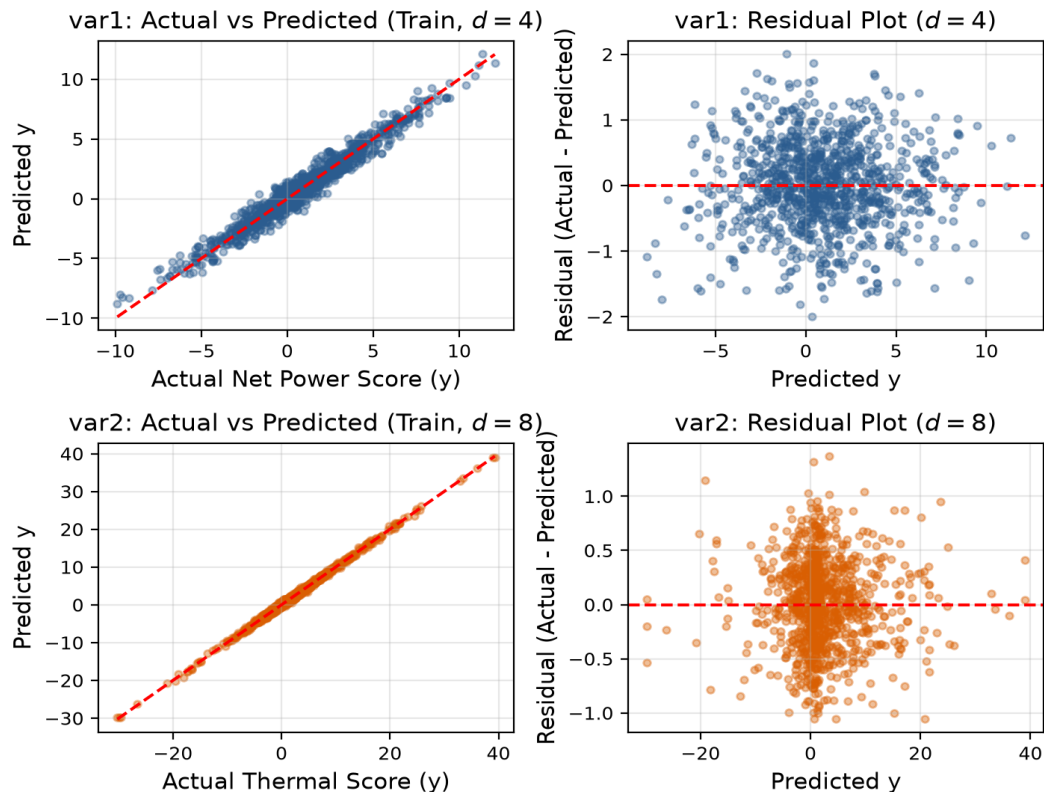


Figure 3: Model diagnostics: (Left) Actual vs. Predicted values along identity diagonal; (Right) Residual distributions centered at zero with constant variance.

Residual Behavior: As evident in Figure 3, residuals for both models exhibit ideal homoscedasticity—evenly distributed around zero across the fitted range without curvature, heteroscedastic fan shapes, or systematic skewness. Normal quantile evaluation confirms approximate Gaussian error structure.

Test Prediction Distribution Check: Predictions on the hidden 1,000-sample test sets display exceptional physical consistency: Phase 1 predictions yield mean = 0.9876, std = 4.1168, range [-10.77, 15.71] (training target: mean 0.8629, std 3.1872, range [-9.88, 12.07]). Phase 2 predictions yield mean = 2.2535, std = 6.4661, range [-25.28, 38.92] (training target: mean 2.2751, std 6.9016, range [-30.26, 39.33]). Zero unbounded polynomial extrapolation artifacts were observed.

6. Summary of Deliverables and Reproducibility

All required deliverables have been compiled and verified in accordance with the assignment guidelines:

Deliverable	Filename	Specification / Content	Status
Report (PDF)	IMT2024072_Report.pdf	Professional 4-page academic submission write-up	Completed
Report (LaTeX)	report.tex	Standalone LaTeX source code	Completed
Prediction 1	IMT2024072_pred_var1.csv	1,000 test predictions (y) for Phase 1 (Degree 4)	Verified (1,000 rows)
Prediction 2	IMT2024072_pred_var2.csv	1,000 test predictions (y) for Phase 2 (Degree 8)	Verified (1,000 rows)
Pipeline Code	polynomial_regression.py	Standalone scikit-learn training & inference script	Reproducible

Table 4: Deliverables summary for Roll Number IMT2024072.

Reproducibility Instructions: The pipeline is self-contained. Running python polynomial_regression.py executes the complete workflow: loads calibration datasets, runs 5-fold cross-validation, fits the optimal polynomial models in *scikit-learn*, generates diagnostic plots, and outputs the final submission prediction CSV files.